Fractions, Decimals and Percentages

Name _____ Date _____ Calculator allowed Version 1 Scan to practise this topic online

KEY WORDS fraction *(hautanga)* · decimal *(tau ā-ira)* · percentage *(ōrau)* · equivalent *(ōrite)* · convert *(panoni)*

A Converting ●○○

Remember: Per cent means "out of 100". To change a decimal to a percentage, multiply by 100.

1. Write each fraction as a decimal.

a) $\frac{3}{4}$ = _____ **b)** $\frac{2}{5}$ = _____ **c)** $\frac{7}{8}$ = _____ **d)** $\frac{3}{20}$ = _____

e) $\frac{1}{8}$ = _____ **f)** $\frac{9}{40}$ = _____ **g)** $\frac{17}{50}$ = _____ **h)** $\frac{5}{16}$ = _____

2. Write as a percentage.

a) 0.35 = _____ **b)** 0.07 = _____ **c)** 1.2 = _____ **d)** $\frac{3}{5}$ = _____

e) $\frac{7}{25}$ = _____ **f)** $\frac{5}{8}$ = _____ **g)** $\frac{9}{20}$ = _____ **h)** 0.004 = _____

3. Write as a fraction in simplest form.

a) 0.6 = _____ **b)** 0.45 = _____ **c)** 12% = _____ **d)** 37.5% = _____

e) 0.125 = _____ **f)** 0.04 = _____ **g)** 85% = _____ **h)** 2.5% = _____

B Ordering ●●○

Remember: Change every number to the same form, such as decimals, before you compare them.

4. Write each set in order, smallest first.

a) $0.6, \quad 58\%, \quad \frac{5}{8}$

b) $\frac{1}{3}, \quad 0.3, \quad 35\%$

c) $72\%, \quad \frac{3}{4}, \quad 0.7$

d) $1.2, \quad 125\%, \quad \frac{9}{8}$

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C Fractions and percentages of amounts ●●○

5. Work out:

a) $\frac{3}{4}$ of 28 **b)** $\frac{2}{5}$ of 45 **c)** $\frac{5}{8}$ of \$72

d) $\frac{7}{10}$ of 250 **e)** $\frac{4}{9}$ of 63 **f)** $\frac{5}{6}$ of 3.6 kg

6. Work out:

a) 35% of 240 **b)** 7.5% of 800 **c)** 2.5% of \$1,600

d) 125% of 48 **e)** 0.5% of 3,000 **f)** 18% of 65 kg

D One amount as a percentage of another ●●○

⚠ **Watch out:** Put the part on top and the whole on the bottom, and make sure both are in the same units.

7. Answer each question.

a) What percentage of 250 is 45? **b)** What percentage of 64 is 24?

c) Write 18 out of 40 as a percentage. **d)** Write \$36 as a percentage of \$150.

e) Write 45 minutes as a percentage of 2 hours. **f)** Write 350 g as a percentage of 2 kg.

E Finding the whole amount ●●●

💡 **Remember:** Find 1% (or one part of the fraction) first, then build up to the whole.

8. Find the number.

a) 20% of a number is 14. **b)** $\frac{3}{4}$ of a number is 36.

c) 35% of a number is 91. **d)** $\frac{5}{8}$ of a number is 45.

e) 12.5% of a number is 6. **f)** $\frac{2}{3}$ of a number is 50.

F Problems ●●●

9. A phone battery holds 4,000 mAh when it is full. It is now at 35%. How much charge is left?

10. Kiri saves 15% of her \$640 weekly pay. How much does she save in 4 weeks?

11. Ana scores 34 out of 40 in a test. What is her mark as a percentage?

12. In a class of 30 students, 40% walk to school and $\frac{1}{3}$ come by bus. The rest come by car. How many come by car?

13. Which is the better mark, 17 out of 20 or 42 out of 50? Show how you know.

14. A 240 g bag of trail mix is $\frac{3}{8}$ nuts. How many grams are not nuts?

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Extension

15. Write $\frac{1}{3}$ as a percentage, correct to 1 decimal place.

16. $\frac{2}{5}$ of a number is 36. What is 75% of the number?

17. A shop sells a jacket for $\frac{3}{4}$ of its usual price. Then it takes another 20% off. What fraction of the usual price is the final price?

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Fractions, Decimals and Percentages

Name _____ Date _____ Calculator allowed Version 2 Scan to practise this topic online

KEY WORDS fraction *(hautanga)* · decimal *(tau ā-ira)* · percentage *(ōrau)* · equivalent *(ōrite)* · convert *(panoni)*

A Converting ●○○

Remember: Per cent means "out of 100". To change a decimal to a percentage, multiply by 100.

1. Write each fraction as a decimal.

a) $\frac{1}{4}$ = _____ **b)** $\frac{3}{5}$ = _____ **c)** $\frac{5}{8}$ = _____ **d)** $\frac{7}{20}$ = _____

e) $\frac{3}{8}$ = _____ **f)** $\frac{11}{40}$ = _____ **g)** $\frac{23}{50}$ = _____ **h)** $\frac{3}{16}$ = _____

2. Write as a percentage.

a) 0.45 = _____ **b)** 0.03 = _____ **c)** 1.5 = _____ **d)** $\frac{4}{5}$ = _____

e) $\frac{9}{25}$ = _____ **f)** $\frac{3}{8}$ = _____ **g)** $\frac{11}{20}$ = _____ **h)** 0.006 = _____

3. Write as a fraction in simplest form.

a) 0.8 = _____ **b)** 0.35 = _____ **c)** 16% = _____ **d)** 62.5% = _____

e) 0.375 = _____ **f)** 0.08 = _____ **g)** 45% = _____ **h)** 7.5% = _____

B Ordering ●●○

Remember: Change every number to the same form, such as decimals, before you compare them.

4. Write each set in order, smallest first.

a) $0.7, \quad 68\%, \quad \frac{3}{4}$

b) $\frac{2}{3}, \quad 0.6, \quad 65\%$

c) $82\%, \quad \frac{4}{5}, \quad 0.85$

d) $1.4, \quad 135\%, \quad \frac{11}{8}$

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C Fractions and percentages of amounts ●●○

5. Work out:

a) $\frac{3}{4}$ of 36 **b)** $\frac{3}{5}$ of 35 **c)** $\frac{3}{8}$ of \$56

d) $\frac{9}{10}$ of 350 **e)** $\frac{5}{9}$ of 72 **f)** $\frac{2}{3}$ of 4.5 kg

6. Work out:

a) 45% of 260 **b)** 2.5% of 600 **c)** 7.5% of \$1,200

d) 150% of 36 **e)** 0.5% of 5,000 **f)** 24% of 75 kg

D One amount as a percentage of another ●●○

⚠ **Watch out:** Put the part on top and the whole on the bottom, and make sure both are in the same units.

7. Answer each question.

a) What percentage of 300 is 54? **b)** What percentage of 96 is 36?

c) Write 27 out of 60 as a percentage. **d)** Write \$42 as a percentage of \$175.

e) Write 36 minutes as a percentage of 3 hours. **f)** Write 450 g as a percentage of 3 kg.

E Finding the whole amount ●●●

💡 **Remember:** Find 1% (or one part of the fraction) first, then build up to the whole.

8. Find the number.

a) 40% of a number is 18. **b)** $\frac{2}{5}$ of a number is 26.

c) 15% of a number is 42. **d)** $\frac{3}{8}$ of a number is 27.

e) 2.5% of a number is 9. **f)** $\frac{5}{6}$ of a number is 45.

F Problems ●●●

9. A phone battery holds 5,000 mAh when it is full. It is now at 45%. How much charge is left?

10. Wiremu saves 12% of his \$750 weekly pay. How much does he save in 4 weeks?

11. Tama scores 51 out of 60 in a test. What is his mark as a percentage?

12. In a class of 28 students, 25% walk to school and $\frac{1}{2}$ come by bus. The rest come by car. How many come by car?

13. Which is the better mark, 21 out of 25 or 33 out of 40? Show how you know.

14. A 320 g bag of trail mix is $\frac{5}{8}$ nuts. How many grams are not nuts?

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Extension

15. Write $\frac{2}{3}$ as a percentage, correct to 1 decimal place.

16. $\frac{3}{5}$ of a number is 48. What is 25% of the number?

17. A shop sells a jacket for $\frac{4}{5}$ of its usual price. Then it takes another 25% off. What fraction of the usual price is the final price?

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